

Name: _____

Date: _____

PERCENTAGES, (PERCENTAGE INCREASE AND DECREASE)

LO: I can increase or decrease an amount by a given percentage using a multiplier.

Percentage increase and decrease

A percentage increase makes an amount larger. A percentage decrease makes it smaller. Use a single multiplier for efficiency.

Multiplier method: Increase by 15% means $\times 1.15$. Decrease by 15% means $\times 0.85$.

Quick examples

●	Increase £80 by 25%: 80×1.25 = £100	multiplier 1.25
●	Decrease £60 by 10%: 60×0.90 = £54	multiplier 0.90
●	VAT at 20%: multiply by 1.20	adding 20%
●	Increase by 25% then decrease by 25% = original	does not return to original

Key idea

To increase, add the percentage to 100% to get the multiplier. To decrease, subtract from 100%.

New = Original $\times$ multiplier

Increase: $1 + \frac{\%}{100}$ **Decrease:** $1 - \frac{\%}{100}$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – percentage increase

Increase £240 by 15%

$$\begin{aligned}\text{Multiplier} &= 1 + 0.15 = 1.15 \\ 240 \times 1.15 \\ &= \text{£276}\end{aligned}$$

= £276

The multiplier 1.15 means $100\% + 15\% = 115\%$ of original.

Example 2 – percentage decrease

Decrease £350 by 20%

$$\begin{aligned}\text{Multiplier} &= 1 - 0.20 = 0.80 \\ 350 \times 0.80 \\ &= \text{£280}\end{aligned}$$

= £280

0.80 means keeping 80% (removing 20%).

Example 3 – VAT calculation

A TV costs £450 before VAT at 20%.
Find the price including VAT.

$$\begin{aligned}\text{Multiplier} &= 1.20 \\ 450 \times 1.20 \\ &= \text{£540}\end{aligned}$$

= £540

VAT is a percentage increase on the original price.

Example 4 – reverse percentage

After a 20% increase, a price is £360.
Find the original.

$$\begin{aligned}\text{£360} &= \text{original} \times 1.20 \\ \text{Original} &= 360 \div 1.20 \\ &= \text{£300}\end{aligned}$$

= £300

Divide by the multiplier to reverse a percentage change.

YOUR TURN – PRACTICE (deliberate practice)

1. Percentage increases

Increase each amount by the given percentage.

- a) £200 by 10%
- b) £450 by 20%
- c) £80 by 25%
- d) £1200 by 5%

2. Percentage decreases

Decrease each amount by the given percentage.

- a) £300 by 10%
- b) £560 by 25%
- c) £90 by 30%
- d) £1500 by 15%

3. Finding the multiplier

Write the single multiplier for each change.

- a) Increase by 35%
- b) Decrease by 12%
- c) Increase by 7.5%
- d) Decrease by 42%

4. Reverse percentage problems

Find the original amount.

- a) After 20% increase, price = £480. Original?
- b) After 10% decrease, value = £540. Original?
- c) After 25% increase, bill = £75. Original?
- d) After 15% decrease, weight = £170. Original?

5. Mixed percentage changes

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To increase by 15%, multiply by 1.15 ☐

Correct answer: _____

Reason: _____

b) To decrease by 20%, multiply by 0.20 ☐

Correct answer: _____

Reason: _____

c) A 20% increase followed by a 20% decrease gives the original ☐

Correct answer: _____

Reason: _____

d) To find original after increase, divide by the multiplier ☐

Correct answer: _____

Reason: _____

e) Decrease by 30% means multiplier is 0.70 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop increases all prices by **20%**, then offers a **20% discount** sale. Explain why the final price is NOT the same as the original. If an item was originally **£50**, what is the sale price?

Expression: _____

Simplified: _____

b) A company makes **£85000** profit after a **12.5% increase** on last year. What was last year's profit? If next year they want **£100000**, what percentage increase from this year's profit do they need (to 1 d.p.)?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

